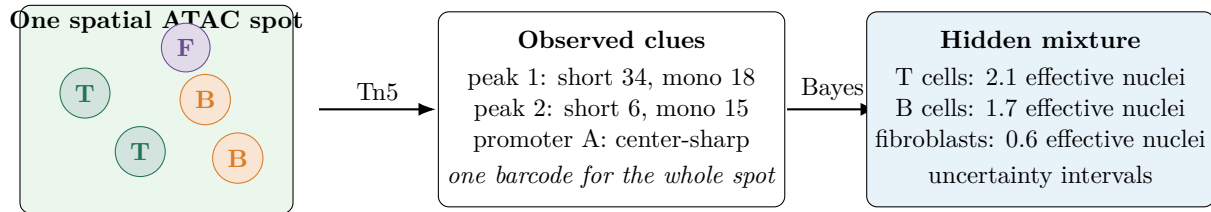


ShapeMix-ATAC

A LaTeX tutorial for the Bayesian fragment-shape model

Written to make the equations, parameters, priors, and hyperparameters easier to read



Main idea

A spatial ATAC spot is treated as a mixture of cell-type chromatin fingerprints. A fingerprint includes not only *which peaks are open*, but also *what the ATAC fragments look like inside those peaks*.

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1 What problem is ShapeMix-ATAC solving?

Spatial ATAC-seq measures open chromatin in a tissue section. The sequencer gives a count of DNA fragments assigned to each spatial spot. A spot may contain more than one nucleus. Therefore, the observed signal is a mixture.

Biology intuition

Think of a spot as a small square drawn on top of tissue. Inside that square, there may be T cells, B cells, fibroblasts, tumor cells, neurons, or other cell types. ATAC-seq does not directly say which nucleus produced each fragment. ShapeMix-ATAC tries to infer the hidden cell mixture from the fragments.

The ordinary deconvolution question is:

$$\text{observed mixed signal} \implies \text{cell-type amounts in the spot.}$$

The ShapeMix-ATAC question is richer:

$$\text{observed mixed ATAC fragment counts and fragment shapes} \implies \text{cell-type amounts with uncertainty.}$$

1.1 Why ATAC needs a special model

Spatial transcriptomics mostly counts RNA molecules from genes. Spatial ATAC counts DNA fragments from open chromatin peaks. ATAC fragments also have physical features:

- **Fragment length:** short fragments often come from nucleosome-free DNA; longer fragments can span nucleosomes.
- **Position inside a peak:** fragments may concentrate near the center, left edge, right edge, or flanking regions.
- **Motif-centered pattern:** cuts around a transcription-factor motif can carry footprint-like information.

Main idea

The novelty of ShapeMix-ATAC is to use a three-dimensional count table:

$$Y_{s,p,b} = \text{number of fragments in spot } s, \text{ peak } p, \text{ shape bin } b.$$

A usual peak-count model uses only $Y_{s,p} = \sum_b Y_{s,p,b}$. ShapeMix keeps the extra index b .

2 The data table: from peak counts to shape counts

2.1 Indices

Symbol	Meaning
$s = 1, \dots, S$	spatial spots or pixels
$p = 1, \dots, P$	ATAC peaks or peak groups
$b = 1, \dots, B$	shape bins, such as short/mono/long fragments
$c = 1, \dots, C$	cell types in the reference, such as T cell, B cell, fibroblast

2.2 Shape-bin examples

A first implementation should use only a few bins. This keeps the method explainable and avoids overfitting.

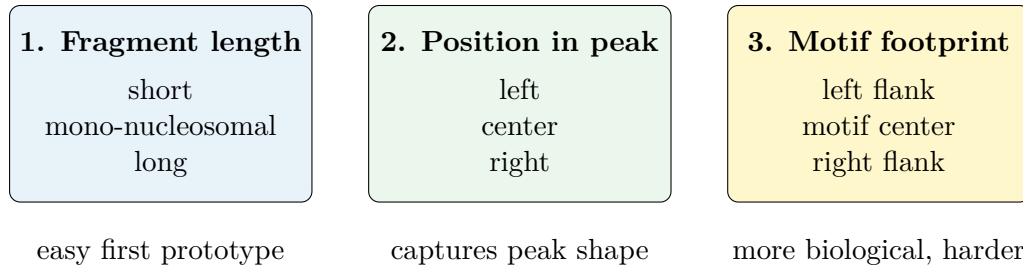


Figure 1: Three possible choices for the shape-bin index b . A first prototype should usually start with fragment-length bins.

For example, with three fragment-length bins:

$$b = 1 : \text{short}, \quad b = 2 : \text{mono-nucleosomal}, \quad b = 3 : \text{long}.$$

3 A short Bayesian warm-up

Bayesian modeling means we write down a story for how the data could have been generated. Then we run the story backward.

Bayes rule in words

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

In ShapeMix-ATAC:

$$p(\text{hidden cell mixture} | \text{observed ATAC fragments}) \propto p(\text{observed ATAC fragments} | \text{hidden cell mixture})p(\text{hidden cell mixture})$$

3.1 Distribution cheat sheet

Distribution	Used for	Intuition
Poisson	simple counts	Counts fluctuate around their mean.
Negative binomial	noisy sequencing counts	Like Poisson, but allows extra noise.
Gamma	positive amounts	Good for abundances, background, dispersion, and rates.
Log-normal	positive multiplicative effects	Good for depth/yield/bias factors.
Beta	probabilities between 0 and 1	Good for peak accessibility probabilities.
Dirichlet	vectors that sum to 1	Good for proportions, such as shape-bin probabilities.
Multinomial	splitting a total count into categories	Good when total nuclei are known and split among cell types.
Normal / CAR normal	real-valued latent spatial fields	Good for smooth spatial variation.

4 Start with the ordinary peak-count model

Before adding shape, an ATAC peak-count model says:

$$Y_{s,p} \sim \text{NegBinomial}(\lambda_{s,p}, \phi_p),$$

where

$$\lambda_{s,p} = \ell_s \left[\beta_p + \sum_{c=1}^C z_{s,c} q_c d_p \theta_{c,p} \right].$$

Read this as a sentence:

The expected ATAC fragments at peak p in spot s equal spot depth ℓ_s , times background plus the summed contributions from each cell type.

The problem is that $Y_{s,p}$ ignores the shapes of fragments inside the peak.

5 The ShapeMix-ATAC likelihood

5.1 The core equation

ShapeMix adds the shape-bin index b :

Core ShapeMix likelihood

$$Y_{s,p,b} \sim \text{NegBinomial}(\lambda_{s,p,b}, \phi_p), \tag{1}$$

$$\lambda_{s,p,b} = \ell_s \left[\beta_{p,b} + \sum_{c=1}^C z_{s,c} q_c d_p \theta_{c,p} \omega_{c,p,b} \right]. \tag{2}$$

This is the central model. The new ShapeMix term is

$$\boxed{\omega_{c,p,b}},$$

which means:

Among fragments from cell type c at peak p , what fraction should fall into shape bin b ?

5.2 Meaning of each factor in $\lambda_{s,p,b}$

Term	Meaning
ℓ_s	sequencing-depth offset for spot s . A deeper spot produces more fragments overall.
$\beta_{p,b}$	background fragments for peak p , shape bin b , not explained by known cell types.
$z_{s,c}$	effective number of cells/nuclei of type c in spot s . This is the main hidden quantity.
q_c	per-cell ATAC yield for cell type c . Some cell types produce more accessible fragments per nucleus.
d_p	peak detectability. Some peaks are easier to observe because of width, GC content, mappability, or Tn5 bias.
$\theta_{c,p}$	accessibility of peak p in cell type c .
$\omega_{c,p,b}$	shape distribution: fraction of fragments from cell type c , peak p , assigned to bin b .
ϕ_p	negative-binomial dispersion for peak p . Smaller ϕ_p means noisier counts.

5.3 Negative-binomial parameterization

This document uses a negative binomial with mean λ and dispersion ϕ :

$$Y \sim \text{NegBinomial}(\lambda, \phi)$$

means

$$\mathbb{E}[Y] = \lambda, \quad \text{Var}(Y) = \lambda + \frac{\lambda^2}{\phi}.$$

A large ϕ makes the model close to Poisson. A small ϕ allows extra sequencing noise.

6 The generative story

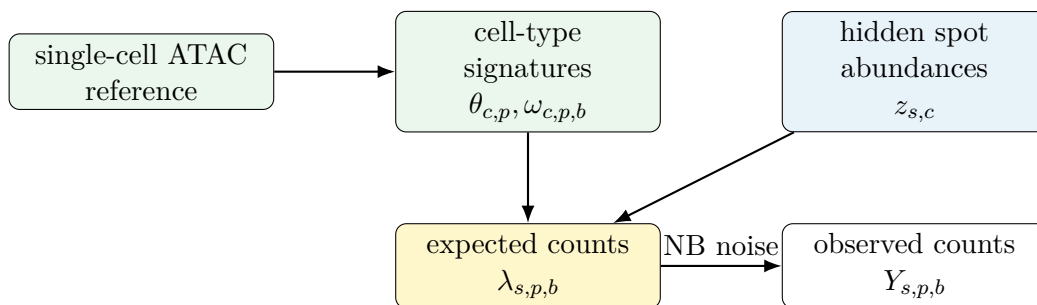


Figure 2: ShapeMix writes down how reference fingerprints and hidden cell abundances generate the observed shape-count tensor.

The model can be taught as a recipe:

1. Learn each cell type's peak accessibility $\theta_{c,p}$ from a scATAC or multiome reference.
2. Learn each cell type's fragment-shape pattern $\omega_{c,p,b}$ from the same reference.
3. Guess a possible hidden cell mixture $z_{s,c}$ for each spot.

4. Use Equation (2) to predict expected counts $\lambda_{s,p,b}$.
5. Compare predicted counts with observed counts $Y_{s,p,b}$.
6. Update the hidden mixture and other uncertain parameters until the posterior is high.

7 Reference signatures

The model needs a reference containing cells with known cell-type labels. A single-cell multiome reference is especially useful because RNA can help label cell types while ATAC provides chromatin signatures.

7.1 Peak accessibility prior

Let $R_{c,p}^{\text{open}}$ and $R_{c,p}^{\text{closed}}$ be reference pseudo-counts for how often peak p appears open or closed in cell type c . Then:

Accessibility prior

$$\theta_{c,p} \sim \text{Beta} \left(a_{\theta} + R_{c,p}^{\text{open}}, b_{\theta} + R_{c,p}^{\text{closed}} \right). \quad (3)$$

Here $\theta_{c,p} \in [0, 1]$. Large $\theta_{c,p}$ means peak p is often accessible in cell type c .

7.2 Shape prior

For each cell type c and peak p , the shape probabilities must sum to 1:

$$\sum_{b=1}^B \omega_{c,p,b} = 1, \quad \omega_{c,p,b} \geq 0.$$

A Dirichlet prior is natural:

Shape prior

$$\boldsymbol{\omega}_{c,p,\cdot} = (\omega_{c,p,1}, \dots, \omega_{c,p,B}), \quad (4)$$

$$\boldsymbol{\omega}_{c,p,\cdot} \sim \text{Dirichlet} \left(a_{\omega} \mathbf{u}_{\omega} + \mathbf{R}_{c,p,\cdot}^{\text{shape}} \right). \quad (5)$$

$\mathbf{R}_{c,p,\cdot}^{\text{shape}}$ is the vector of reference shape-bin counts. $\mathbf{u}_{\omega}$ is a default shape vector, such as uniform $(1/B, \dots, 1/B)$.

Biology intuition

If a T cell reference says that fragments at peak p are usually short, then $\omega_{T,p,\text{short}}$ should be high. If a B cell reference says the same peak mostly has mononucleosomal fragments, then $\omega_{B,p,\text{mono}}$ should be high. ShapeMix uses this difference as extra evidence.

8 Priors for hidden cell abundance

The main hidden quantity is $z_{s,c}$, the effective amount of cell type c in spot s . It is positive:

$$z_{s,c} > 0.$$

A useful prior is a Gamma distribution:

Abundance prior

$$\log \mu_{s,c} = \eta_c + u_{s,c}, \tag{6}$$

$$z_{s,c} \sim \text{Gamma} \left(r_z, \frac{r_z}{\mu_{s,c}} \right). \tag{7}$$

This Gamma parameterization uses shape r_z and rate $r_z/\mu_{s,c}$, so

$$\mathbb{E}[z_{s,c}] = \mu_{s,c}.$$

8.1 What are η_c and $u_{s,c}$?

- η_c is the global baseline abundance of cell type c . Common cell types can have larger η_c .
- $u_{s,c}$ is a spatial effect. It lets a cell type be enriched in some tissue regions and depleted in others.

A simple prior for η_c is:

$$\eta_c \sim \text{Normal}(m_{\eta,c}, \sigma_\eta^2).$$

8.2 Optional spatial prior

Neighboring spots often have related cell mixtures. To encode this, use a conditional autoregressive-like prior:

Spatial smoothing prior

$$\mathbf{u}_{\cdot,c} \sim \text{Normal} \left(\mathbf{0}, \tau_c^{-1} (D - \varphi_c W + \epsilon I)^{-1} \right). \tag{8}$$

Here:

- W is a spot-neighbor graph. $W_{s,t}$ is large if spots s and t are neighbors that should be smoothed together.
- D is diagonal with $D_{s,s} = \sum_t W_{s,t}$.
- τ_c controls how strongly the spatial effect is shrunk toward smoothness.
- φ_c controls spatial dependence. If $\varphi_c = 0$, there is little spatial sharing. If φ_c is large, nearby spots tend to be similar.
- ϵI is a tiny stabilizer that makes the matrix invertible.

For a first prototype, this spatial prior can be removed. Then set $u_{s,c} = 0$, and let each spot be deconvolved independently.

9 Complete core model

Putting the pieces together:

Complete core ShapeMix-ATAC model

$$Y_{s,p,b} \sim \text{NegBinomial}(\lambda_{s,p,b}, \phi_p), \quad (9)$$

$$\lambda_{s,p,b} = \ell_s \left[\beta_{p,b} + \sum_{c=1}^C z_{s,c} q_c d_p \theta_{c,p} \omega_{c,p,b} \right], \quad (10)$$

$$z_{s,c} \sim \text{Gamma} \left(r_z, \frac{r_z}{\exp(\eta_c + u_{s,c})} \right), \quad (11)$$

$$\eta_c \sim \text{Normal}(m_{\eta,c}, \sigma_{\eta}^2), \quad (12)$$

$$\mathbf{u}_{\cdot,c} \sim \text{Normal} \left(\mathbf{0}, \tau_c^{-1} (D - \varphi_c W + \epsilon I)^{-1} \right), \quad (13)$$

$$\tau_c \sim \text{Gamma}(a_{\tau}, b_{\tau}), \quad (14)$$

$$\varphi_c \sim \text{Beta}(a_{\varphi}, b_{\varphi}) \quad \text{truncated to } [0, 1], \quad (15)$$

$$q_c \sim \text{LogNormal}(m_q, \sigma_q^2), \quad (16)$$

$$d_p \sim \text{LogNormal}(m_d + \gamma_w \log \text{width}_p + \gamma_{GC} \text{GC}_p, \sigma_d^2), \quad (17)$$

$$\theta_{c,p} \sim \text{Beta}(a_{\theta} + R_{c,p}^{\text{open}}, b_{\theta} + R_{c,p}^{\text{closed}}), \quad (18)$$

$$\omega_{c,p,\cdot} \sim \text{Dirichlet}(a_{\omega} \mathbf{u}_{\omega} + \mathbf{R}_{c,p,\cdot}^{\text{shape}}), \quad (19)$$

$$\beta_{p,b} \sim \text{Gamma}(a_{\beta}, b_{\beta}), \quad (20)$$

$$\phi_p \sim \text{Gamma}(a_{\phi}, b_{\phi}). \quad (21)$$

The output cell-type proportion is not a new random variable. It is computed from the inferred abundances:

$$\pi_{s,c} = \frac{z_{s,c}}{\sum_{c'=1}^C z_{s,c'}}.$$

10 Parameter guide: what is a hyperparameter?

Rule used in this document

A **hyperparameter** is fixed before inference, chosen by the user, cross-validation, weak default settings, or reference-based pseudo-counts. A **model parameter** or **latent variable** is inferred from the data and has a prior distribution.

10.1 Observed data and fixed inputs

These are not inferred. They are measured or precomputed.

Table 1: Observed data and fixed inputs.

Symbol	Meaning	Hyper?	Distribution / status
$Y_{s,p,b}$	observed spatial ATAC count for spot s , peak p , bin b	No; observed data	Modeled by $\text{NegBinomial}(\lambda_{s,p,b}, \phi_p)$.
ℓ_s	spot sequencing-depth offset	No; fixed offset	Usually computed from total fragments, e.g. total fragments divided by median total fragments. Optional version: put a LogNormal or Gamma prior on it.
$R_{c,p}^{\text{open}}$	reference accessibility pseudo-counts	No; reference data	Used inside the Beta prior for $\theta_{c,p}$.
$R_{c,p}^{\text{closed}}$	reference closed pseudo-counts	No; reference data	Used inside the Beta prior for $\theta_{c,p}$.
$\mathbf{R}_{c,p,\cdot}^{\text{shape}}$	reference shape-bin counts	No; reference data	Used inside the Dirichlet prior for $\omega_{c,p,\cdot}$.
width_p	width of peak p	No; fixed covariate	Used to predict peak detectability d_p .
GC_p	GC content of peak p	No; fixed covariate	Used to predict peak detectability d_p .
$W_{s,t}$	neighbor weight between spots s and t	No; fixed graph	Usually built from spatial coordinates and optional chromatin similarity.
D	diagonal graph-degree matrix	No; fixed matrix	$D_{s,s} = \sum_t W_{s,t}$.
I	identity matrix	No; fixed matrix	Used only for numerical stabilization.

10.2 Inferred model parameters and latent variables

These are not hyperparameters. They are inferred and each follows a prior distribution or a deterministic rule.

Table 2: Inferred quantities and their distributions.

Symbol	Meaning	Hyper?	Distribution / rule
$\lambda_{s,p,b}$	expected count for spot s , peak p , bin b	No	Deterministic function of Equation (2). It does not need its own prior.
$z_{s,c}$	effective amount of cell type c in spot s	No	$z_{s,c} \sim \text{Gamma}(r_z, r_z / \exp(\eta_c + u_{s,c}))$.
$\pi_{s,c}$	cell-type proportion in spot s	No	Deterministic: $\pi_{s,c} = z_{s,c} / \sum_{c'} z_{s,c'}$.
η_c	global baseline abundance for cell type c	No	$\eta_c \sim \text{Normal}(m_{\eta,c}, \sigma_{\eta}^2)$.
$u_{s,c}$	spatial abundance effect for spot s , cell type c	No	$\mathbf{u}_{\cdot,c} \sim \text{Normal}(0, \tau_c^{-1}(D - \varphi_c W + \epsilon I)^{-1})$.
τ_c	spatial precision for cell type c	No	$\tau_c \sim \text{Gamma}(a_{\tau}, b_{\tau})$.

Symbol	Meaning	Hyper?	Distribution / rule
φ_c	spatial dependence strength for cell type c	No	$\varphi_c \sim \text{Beta}(a_\varphi, b_\varphi)$, truncated to $[0, 1)$.
q_c	per-cell ATAC yield for cell type c	No	$q_c \sim \text{LogNormal}(m_q, \sigma_q^2)$.
d_p	peak detectability / technical capture factor	No	$d_p \sim \text{LogNormal}(m_d + \gamma_w \log \text{width}_p + \gamma_{GC} \text{GC}_p, \sigma_d^2)$.
γ_w	coefficient for peak width effect on detectability	No	$\gamma_w \sim \text{Normal}(0, \sigma_\gamma^2)$, or fixed to 0 in the simplest version.
γ_{GC}	coefficient for GC effect on detectability	No	$\gamma_{GC} \sim \text{Normal}(0, \sigma_\gamma^2)$, or fixed to 0 in the simplest version.
$\theta_{c,p}$	accessibility probability/intensity of peak p in cell type c	No	$\theta_{c,p} \sim \text{Beta}(a_\theta + R_{c,p}^{\text{open}}, b_\theta + R_{c,p}^{\text{closed}})$.
$\omega_{c,p,b}$	probability that fragments from (c, p) fall in shape bin b	No	$\omega_{c,p,\cdot} \sim \text{Dirichlet}(a_\omega \mathbf{u}_\omega + \mathbf{R}_{c,p,\cdot}^{\text{shape}})$.
$\beta_{p,b}$	background rate for peak p , bin b	No	$\beta_{p,b} \sim \text{Gamma}(a_\beta, b_\beta)$.
ϕ_p	negative-binomial dispersion for peak p	No	$\phi_p \sim \text{Gamma}(a_\phi, b_\phi)$.

10.3 Hyperparameters

These are fixed before inference. They do not have distributions inside this model unless you decide to build a deeper hierarchical model.

Table 3: Hyperparameters and how to choose them.

Symbol	Meaning	Hyper?	Typical choice
r_z	concentration of Gamma prior on cell abundance	Yes	Start with 2 to 10. Larger means less variability around $\mu_{s,c}$.
$m_{\eta,c}$	prior mean for global abundance of cell type c	Yes	Based on expected cell prevalence, or 0 for weak default.
σ_η	prior standard deviation for η_c	Yes	1 to 2 on log scale.
a_τ, b_τ	Gamma prior parameters for spatial precision τ_c	Yes	Weak positive values, e.g. $a_\tau = 2, b_\tau = 1$.
a_φ, b_φ	Beta prior parameters for spatial dependence φ_c	Yes	Use (2, 2) for neutral or (3, 1.5) if expecting smooth tissue zones.
m_q, σ_q	prior mean and SD for cell-type ATAC yield q_c	Yes	Set $m_q = 0, \sigma_q = 0.5$ or 1 on log scale.
m_d, σ_d	prior mean and SD for peak detectability d_p	Yes	Set $m_d = 0, \sigma_d = 0.5$ or learn from reference QC.
σ_γ	prior SD for detectability covariate coefficients	Yes	0.5 to 1 on log scale.

Symbol	Meaning	Hyper?	Typical choice
a_θ, b_θ	pseudo-counts for accessibility prior	Yes	Small values, e.g. 0.5, 0.5, or stronger if reference labels are trusted.
a_ω	strength of default shape prior	Yes	0.5 to 5. Smaller lets reference shape counts dominate.
$\mathbf{u}_\omega$	default shape vector	Yes	Uniform vector $(1/B, \dots, 1/B)$, or global ATAC shape distribution.
a_β, b_β	Gamma prior parameters for background	Yes	Choose to prefer small background, e.g. mean a_β/b_β near 0.001 to 0.01.
a_ϕ, b_ϕ	Gamma prior parameters for NB dispersion	Yes	Weak positive values, or empirical from peak-level variance.
ϵ	numerical stabilizer in spatial covariance	Yes	Very small, e.g. 10^{-5} .

11 Worked example: why shape helps

Suppose there are only two cell types: T cells and B cells. At one peak, both cell types have the same total accessibility, so a peak-count-only model cannot distinguish them.

Cell type / spot	Total count at peak	Short fragments	Mono fragments
T-cell reference	100	80%	20%
B-cell reference	100	20%	80%
Observed spot	100	60%	40%

Peak-count model sees only:

$$100 \approx 100.$$

It learns almost nothing.

ShapeMix sees:

$$0.60 \approx 0.80\pi_T + 0.20(1 - \pi_T).$$

Solving:

$$0.60 = 0.80\pi_T + 0.20 - 0.20\pi_T,$$

$$0.40 = 0.60\pi_T,$$

$$\pi_T \approx 0.67.$$

So the observed spot looks like roughly:

67% T cell	+	33% B cell.
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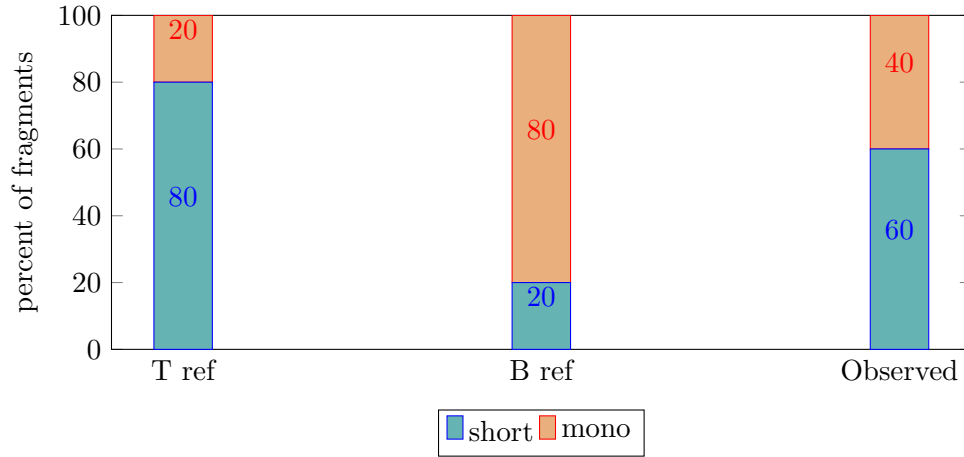


Figure 3: The total peak count is identical, but the shape distribution reveals a mixture.

12 What the posterior means

The posterior is the full probability distribution over all unknown quantities after seeing the data:

$$p(\{z, \eta, u, \tau, \varphi, q, d, \gamma, \theta, \omega, \beta, \phi\} | Y) \quad (22)$$

$$\propto \prod_{s,p,b} p(Y_{s,p,b} | \lambda_{s,p,b}, \phi_p) \prod_{s,c} p(z_{s,c} | \eta_c, u_{s,c}) \prod_c p(\eta_c) p(\mathbf{u}_{\cdot,c} | \tau_c, \varphi_c) p(\tau_c) p(\varphi_c) \quad (23)$$

$$\times \prod_c p(q_c) \prod_p p(d_p) \prod_{c,p} p(\theta_{c,p}) p(\omega_{c,p,\cdot}) \prod_{p,b} p(\beta_{p,b}) \prod_p p(\phi_p). \quad (24)$$

The model output should report not only one best estimate, but also uncertainty:

$$\hat{\pi}_{s,c} = \mathbb{E}[\pi_{s,c} | Y], \quad 95\% \text{ credible interval for } \pi_{s,c}.$$

Why uncertainty matters

If a rare cell type is predicted at 2%, but the 95% credible interval is 0% to 12%, the model is not confident. This is better than pretending the 2% estimate is exact.

13 Fitting the model

There are two practical inference choices.

13.1 Option 1: MAP estimation

MAP means maximum a posteriori. It finds the single parameter setting that maximizes:

$$\log p(Y | \Theta) + \log p(\Theta),$$

where Θ means all unknown quantities.

This is easier to implement and debug. It is a good first prototype.

13.2 Option 2: variational inference

Variational inference approximates the posterior with a simpler distribution:

$$q(\Theta) \approx p(\Theta | Y).$$

Then it optimizes an evidence lower bound, usually called the ELBO:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_q[\log p(Y, \Theta)] - \mathbb{E}_q[\log q(\Theta)].$$

This gives uncertainty estimates and scales better than MCMC for many spots and peaks.

Recommended development path

Start with MAP for the minimal model. Once it works, implement variational inference for uncertainty and scalability.

14 Minimal prototype

The full model is useful for a paper, but the first code version should be simpler.

14.1 Minimal model equations

Fix $\theta_{c,p}$, $\omega_{c,p,b}$, d_p , q_c , and ϕ_p from reference or empirical estimates. Infer only $z_{s,c}$ and maybe $\beta_{p,b}$:

$$Y_{s,p,b} \sim \text{NegBinomial}(\lambda_{s,p,b}, \phi_p), \quad (25)$$

$$\lambda_{s,p,b} = \ell_s \left[\beta_{p,b} + \sum_c z_{s,c} q_c d_p \theta_{c,p} \omega_{c,p,b} \right], \quad (26)$$

$$z_{s,c} \sim \text{Gamma}(r_z, r_z/m_c), \quad (27)$$

$$\beta_{p,b} \sim \text{Gamma}(a_\beta, b_\beta). \quad (28)$$

14.2 Minimal parameter table

Symbol	Hyper?	Distribution / status
$Y_{s,p,b}$	No; observed	$Y_{s,p,b} \sim \text{NegBinomial}(\lambda_{s,p,b}, \phi_p)$.
$z_{s,c}$	No	$z_{s,c} \sim \text{Gamma}(r_z, r_z/m_c)$.
$\beta_{p,b}$	No	$\beta_{p,b} \sim \text{Gamma}(a_\beta, b_\beta)$.
$\theta_{c,p}$	No; fixed estimate	Estimated from reference and treated as fixed in first prototype.
$\omega_{c,p,b}$	No; fixed estimate	Estimated from reference and treated as fixed in first prototype.
q_c, d_p, ϕ_p, ℓ_s	No; fixed estimates/offsets	Empirical estimates.
$r_z, m_c, a_\beta, b_\beta$	Yes	Fixed hyperparameters.

This minimal model is already ShapeMix because it uses $\omega_{c,p,b}$.

15 How to build the shape-count tensor

15.1 Step-by-step preprocessing

1. Start with an ATAC fragment file containing chromosome, start, end, and spatial barcode.
2. Intersect each fragment with a chosen peak set.
3. Assign each fragment to a shape bin.
4. Count fragments by spot, peak, and shape bin.
5. Keep informative peaks or peak groups.
6. Build the tensor $Y_{s,p,b}$.

15.2 Fragment-length bin example

For fragment length L :

$$b(L) = \begin{cases} 1, & L < 100 \quad \text{short,} \\ 2, & 100 \leq L < 250 \quad \text{mono-nucleosomal-like,} \\ 3, & L \geq 250 \quad \text{long.} \end{cases}$$

The exact cutoffs should be tuned to the assay and species.

16 Diagnostics and ablation tests

16.1 Diagnostics

After fitting, check:

- **Posterior predictive checks:** simulate Y^{rep} from the fitted model and compare with observed Y .
- **Shape-bin reconstruction:** does the model reproduce short/mono/long ratios?
- **Rare-cell uncertainty:** rare cell types should have wider intervals unless evidence is strong.
- **Reference mismatch:** large background $\beta_{p,b}$ or systematic residuals suggest missing cell states.
- **Spatial smoothness:** if using the spatial prior, check that it smooths within tissue domains but does not erase boundaries.

16.2 Ablation tests

Ablation means removing one ingredient at a time to test whether it matters.

Model version	Question answered
Peak-count only	Does shape add value beyond total accessibility?
ShapeMix with length bins	Does fragment length improve deconvolution?
ShapeMix with position bins	Does within-peak position improve deconvolution?
No spatial prior	Does spatial smoothing help?
No peak detectability d_p	Are peak-specific technical effects important?
No background $\beta_{p,b}$	Is unexplained signal being forced into wrong cell types?
Fixed signatures vs Bayesian signatures	Does uncertainty in the reference matter?

17 What makes ShapeMix-ATAC new and testable?

The standard transferred deconvolution methods use spot-by-feature count matrices. For ATAC, the feature is usually a peak. ShapeMix changes the data object from a matrix to a tensor:

$$\text{ordinary ATAC deconvolution: } Y_{s,p} \implies \text{ShapeMix-ATAC: } Y_{s,p,b}.$$

Main idea

ShapeMix-ATAC tests the hypothesis that fragment-shape information helps distinguish cell types whose total accessibility profiles are too similar.

This is especially useful for:

- closely related immune subtypes,
- nearby developmental states,
- tumor and stromal mixtures,
- rare cell types with subtle peak-count differences,
- cell types that share open peaks but differ in nucleosome or footprint patterns.

18 Implementation pseudocode

Pseudocode

Input:

```
spatial ATAC fragments
peaks or peak groups
scATAC/multiome reference with cell type labels
spatial coordinates
```

Preprocess:

```
assign each spatial fragment to a peak p
assign each fragment to a shape bin b
build tensor Y[s,p,b]
build reference signatures theta[c,p] and omega[c,p,b]
```

Fit minimal model:

```
initialize z[s,c] using peak-count deconvolution or NNLS
repeat until convergence:
    compute lambda[s,p,b]
    update z[s,c] to improve NB likelihood plus prior
    optionally update beta[p,b]
```

Fit full model:

```
add priors for theta, omega, q, d, phi
add spatial prior for u[s,c]
use variational inference or MAP optimization
```

Output:

```
pi[s,c] = z[s,c] / sum_c z[s,c]
uncertainty intervals
posterior predictive checks
residual peaks and residual shape patterns
```

19 Common questions

19.1 Is $\omega_{c,p,b}$ a hyperparameter?

No. In the full Bayesian model, $\omega_{c,p,b}$ is an inferred model parameter with a Dirichlet prior. In the minimal prototype, it can be fixed from the reference. If fixed, it is not a hyperparameter in the usual Bayesian sense; it is an empirical input.

19.2 Is $\theta_{c,p}$ a hyperparameter?

No. In the full model, $\theta_{c,p}$ is an inferred model parameter with a Beta prior. The hyperparameters are a_θ , b_θ , and the reference pseudo-count construction.

19.3 Is $z_{s,c}$ a cell count?

It is an *effective* cell count. It behaves like a positive abundance. It may not be an exact integer because spatial ATAC counts depend on capture efficiency, chromatin openness, and sequencing depth. If image-derived nuclei counts are available, the model can constrain $\sum_c z_{s,c}$ to match them.

19.4 Can we use Poisson instead of negative binomial?

Yes. The Poisson version is:

$$Y_{s,p,b} \sim \text{Poisson}(\lambda_{s,p,b}).$$

It is simpler and may work well for some ATAC peak counts. The negative binomial is safer when the data are overdispersed.

19.5 Do we need all peaks?

No. Use selected peaks or peak groups. ShapeMix is most useful when the selected peaks have informative shape differences between cell types. A practical first version might use 5,000 to 30,000 peaks or groups, depending on depth.

20 Optional extension: image-based cell budget

If nuclei segmentation gives an approximate number of nuclei N_s^{img} in spot s , add:

$$N_s^{\text{img}} \sim \text{Normal}_+ \left(\sum_c z_{s,c}, \sigma_N^2 \right),$$

where Normal_+ means a Normal distribution truncated to positive values.

Parameter status:

Symbol	Hyper?	Distribution / status
N_s^{img}	No; observed	Image-derived nuclei count.
σ_N	Yes	Fixed measurement-error scale for image-derived nuclei counts.
$\sum_c z_{s,c}$	No	Deterministic total effective cell count in spot s .

21 Optional extension: peak reliability weights

Some peaks are noisy, poorly mappable, or not useful for deconvolution. Add a reliability weight r_p :

$$r_p \sim \text{Beta}(a_r, b_r), \quad 0 < r_p < 1.$$

Then use a weighted log-likelihood:

$$\mathcal{L}_{\text{weighted}} = \sum_{s,p,b} r_p \log p(Y_{s,p,b} | \lambda_{s,p,b}, \phi_p).$$

Parameter status:

Symbol	Hyper?	Distribution / status
r_p	No	$r_p \sim \text{Beta}(a_r, b_r)$.
a_r, b_r	Yes	Fixed Beta prior parameters.

This extension is useful, but it is not required for the first ShapeMix paper. The core novelty is already the shape-count likelihood.

22 Recommended first experiment

1. Build a peak-count baseline using $Y_{s,p} = \sum_b Y_{s,p,b}$.
2. Build ShapeMix with two or three fragment-length bins.
3. Simulate pseudo-spots from a scATAC or multiome reference.
4. Compare true and estimated $\pi_{s,c}$.
5. Stress-test related cell types and rare cell types.
6. Report whether shape improves RMSE, Jensen-Shannon divergence, rare-cell F1, and calibration.

Important caution

Do not begin with too many shape bins. If P peaks and B bins are both large, the tensor becomes sparse. Start simple, prove shape helps, then add richer shape features.

23 Summary in one page

The whole model in one line

$$Y_{s,p,b} \sim \text{NegBinomial} \left(\ell_s \left[\beta_{p,b} + \sum_c z_{s,c} q_c d_p \theta_{c,p} \omega_{c,p,b} \right], \phi_p \right).$$

- $Y_{s,p,b}$: observed shape-count tensor.
- $z_{s,c}$: hidden amount of cell type c in spot s .
- $\theta_{c,p}$: how open peak p is in cell type c .
- $\omega_{c,p,b}$: what shape fragments from cell type c have at peak p .
- d_p : peak detectability.
- q_c : cell-type ATAC yield.
- $\beta_{p,b}$: background.
- ϕ_p : overdispersion.

The final cell-type composition is:

$$\pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.$$

Main idea

ShapeMix-ATAC is easy to explain because it says: *use the pattern of fragments, not only the number of fragments*. It is intuitive because ATAC fragments physically reflect chromatin structure. It is testable because you can compare the same model with and without the shape-bin index b .

References and positioning notes

References

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